

Internet Technology

26FS

Programme	BSc in Business Information Technology
Degree	Bachelor
ECTS	5
Module type	mandatory module
Module coordinator	Dr. Devid Montecchiari
Compulsory attendance	Attendance is expected
Leading principle / Short description	This module gives students an overview of the most important methods and concepts of a web architecture and focuses on the competent use of selected and state-of-the-art web technologies, which are relevant in an enterprise application domain. Besides, this module drives the digitalization by reflecting concepts such as Web APIs, API Lifecycle Management, microservice design and web technologies. Finally, this module sharpens the understanding of an economical software development using frameworks and selected libraries as well as deployment scenarios.
Module content	API Lifecycle Management - Types of APIs, strategies for developing and managing APIs - API development platforms and tools - OpenAPI Web Technologies - REST Architecture and APIs - Development in the Cloud - AI pair programming Web Frontend Frameworks/Libraries - Low-code/No-code approach for front-end development (HTML, CSS, JavaScript) Web Applications - Microservices - Web Authentication - Web Frameworks - Persistence Layer: Object-Relational Mapping - Business and Service Layer: Business Services and REST Controllers - Cloud Deployment
Competencies to be achieved	Knowledge and Understanding: <i>Students ...</i> - know and understand the terms and concepts of a web architecture. - know different diagram types to model and describe web architectures. - know a current (Web) technology to digitise business processes. - know the potential of different methods of web engineering Applying Knowledge and Understanding: <i>Students ...</i> - can describe the use of web technologies using standardised methods and tools. - can analyse and implement web applications using APIs. - can describe and explain the significance of web technologies in practice. Making Judgements: <i>Students ...</i> - are able to model a web application and design a corresponding architecture adequately. - are able to analyse, interpret and describe requirements for a web application. Communications Skills: <i>Students ...</i> - can explain the most critical tasks and activities of web engineering. Learning Skills: <i>Students ...</i> - can develop technical concepts with the help of written tutorials and framework documentation.
Prerequisites	Programming Database Technology
Teaching and learning methods	Contact Hours: - Lecture, Interactive instruction, Discussion, Assignment, Group work Guided Self-Study: - Individual work, Group work, Online Tutorial
Literature	Required Reading: Will be announced during the lecture

**Grading
Assessment**

Grade 1-6 (half grades)

Exam 50%

Oral / Written	Written
Duration (min)	45
Timeframe	Examination period (end of semester)
Grading Scale	points
Remarks	The individual written exam takes place during the official exam period and must be done using the FHNW School of Business iPad/Moodle exam infrastructure and system. The iPad/Moodle exam is graded with a mark between 1 and 6 and counts 50% to the overall module grade.

Project 50%

Timeframe	during the semester (final week of the semester)
Grading Scale	points
Remarks	Group Work Assessed will be the demonstrator (implemented workflow), a demonstration/presentation/video and an online report (for details see assessment form). The submission date may be in the final week of the semester, and this group project leads to a group grade. If a person's participation is significantly different from the group's contribution, the evaluators may provide a separate evaluation.

Module details**Internet Technology (BITPT23) - Fri - Windisch**

Time	1:15 PM - 5:00 PM
Language	Englisch
Periodicity	Weekly
Lecturers	Dr. Devid Montecchiari
Number	1-26FS.W-B-BIT-060305S1.SN/PT23

Internet Technology (BITFT 2a & BITPT 3b) - Wed - Basel

Time	1:15 PM - 5:00 PM
Language	Englisch
Max. participants	48
Periodicity	Weekly
Lecturers	Dr. Devid Montecchiari
Number	2-26FS.W-B-BIT-040305S1.SN/FT24PT23

